

Problem 1: House Price Prediction using Gradient Descent

Dataset:

House Prices Dataset (Kaggle)

Tasks

1. Load the dataset using Pandas.
 2. Select one feature (e.g., area) and the target variable (price). And convert them to numpy array
 3. Implement the Mean Squared Error (MSE) Cost Function manually.
 4. Calculate the initial cost using random values of slope and intercept.
 5. Apply Gradient Descent from scratch.
 6. Train for 1000 iterations.
 7. Plot:
 - Iteration vs Cost
 - Actual vs Predicted Values
 8. Report the final learned parameters.
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Problem 2: Student Performance Prediction

Dataset:

Student Performance Dataset

Tasks

1. Predict math score using reading score. convert them to numpy array from the dataframe
 2. Implement the MSE Cost Function manually.
 3. Compute the gradient formulas manually.
 4. Perform Gradient Descent for 500 iterations.
 5. Print the cost after every 50 iterations.
 6. Plot Cost vs Iteration.
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Problem 3: Insurance Charges Prediction

Dataset:

Medical Cost Personal Dataset

Tasks

1. Use age as input feature.
2. Use charges as target.
3. Convert them to numpy array from the dataframe
4. Implement:
 - Cost Function
 - Gradient Calculation

- Gradient Descent
 - 5. Train for 1000 iterations.
 - 6. Compare initial and final costs.
 - 7. Visualize the regression line.
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Problem 4: Salary Prediction

Dataset:

Salary Dataset

Tasks

1. Predict Salary from YearsExperience.
 2. Initialize parameters randomly.
 3. Calculate cost before training.
 4. Apply Gradient Descent.
 5. Plot:
 - Cost vs Iteration
 - Regression Line
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Problem 5: Car Fuel Consumption Prediction

Dataset:

Auto MPG Dataset

Tasks

1. Use horsepower to predict mpg.
 2. Implement MSE Cost Function manually.
 3. Apply Gradient Descent.
 4. Train for different learning rates:
 - 0.001
 - 0.01
 - 0.1
 5. Compare the convergence for different learning rates.
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